MIL-G-27693A(EL)
AMENDMENT 3
13 December 1967
SUPERSEDING
AMENDMENT 2
14 May 1965

MILITARY SPECIFICATION

GYROSCOPE, DISPLACEMENT SBU-6/A, ROLL AND PITCH

This amendment forms a part of Military Specification MIL-G-27693A(USAF) dated 25 September 1962.

Page 2, paragraph 2.1: Under "Standards, Military", add:

"MIL-STD-704 Electric Power, Aircraft Characteristics and Utilization Of"

Page 2, paragraph 3.1: Delete and substitute:

"3.1 Qualification. The gyroscope furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list."

Page 4, Figure 1: Delete and substitute the attached Figure 1.

Page 5, paragraph 3.5.5, add: "When operated at 320 to 350 cps, the maximum free drift shall be not greater than 31° per hour."

Page 7, paragraph 3.8, line 3: Delete "of 4° ± 1.5°" and substitute "not exceeding 5.5°".

Page 8, table I: Delete and substitute the attached table I. Table I forms page 2 of this amendment.

Page 9, paragraph 3.10f, line 3: Delete "V" and substitute "U".

Page 10, paragraph 3.14: In first sentence, add: "in accordance with MIL-STD-704". In line 4, delete "350 to 420 cps" and substitute "320 to 480 cps".

Page 13, paragraph 4.2a: Delete "Preproduction" and substitute "Qualification".

Page 14, paragraphs 4.4, 4.4.1, and 4.4.2: Delete "preproduction" wherever referenced and substitute "qualification."

FSC 6615

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"TABLE I. Pin Connections

Pin	FUNCTION
A	115/200V, 400 cps
B	115/200V, 400 cps
C	115/200V, 400 cps
D	Neutral of 4-wire 3-phase Y, 400 cps
U x V y T z	Roll synchro output
P x N y M z	Pitch synchro output
Y	Power warning flag
H	Pitch erection cutout
K	Roll erection cutout
J	Normal erection running and starting (when connected to pins H and K)
F	Manual fast erection (when connected to pins H and K)
S R	Roll synchro excitation (R is ground)
L G	Pitch synchro excitation (G is ground)
a b c	Pitch external torquing (pin a connected to pin b for normal running and starting; pin a connected to pin c to precess top forward)
W Z E	Roll external torquing (pin W connected to pin Z for normal running and starting; pin W connected to pin E to precess top to the right)
X	Spare

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Page 16, paragraph 4.5.3: Delete test groups I, II, and III and substitute:

"Group I	Group II	Group III
Radio noise interference		Life
High temperature exposure	Salt spray	Vibration failure
Low temperature-high altitude	Humidity	Acceleration."
Thermal shock	Fungus	
Dielectric strength	Rain	
	Sand and dust	

Page 18, paragraph 4.6.8, third sentence: Delete and substitute: "With proper correction for the earth's rotation, the computed drift rate in either axis shall not exceed 25° per hour, except that when the frequency is between 320 and 350 cps, the maximum free drift shall not exceed 31° per hour."

Page 19, paragraph 4.6.15: Delete and substitute:

"4.6.15 Low and high voltage and frequency operation. Power shall be applied to the gyroscope and tests run at the following combinations of voltage and frequency. The gyroscope shall then meet the tests specified in 4.6.8, 4.6.12, and 4.6.13.

a. 195V 320 cps

b. 195V 480 cps

c. 214V 320 cps

d. 214V 480 cps."

Page 20, paragraph 4.6.16d: Delete and substitute:

"d. The no-load output shall be 11.9 ± 0.6V and the no-load phase shall be 3.5° ± 1.5°. The voltage drop from the nominal no-load output shall not exceed 1.2V for the five loads (rotors open), but under no circumstances shall the output voltage be less than 10.7 volts for the five loads (rotors open). The phase shift with five loads (rotors open) shall not exceed a 2.5° increase in phase shift from the no-load value."

Page 24, paragraph 6.4: Delete and substitute:

"6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this

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requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

The activity responsible for the Qualified Products List is the US Army Electronics Command, ATTN: AMSEL-PP-EA, Fort Monmouth, New Jersey 07703, and information pertaining to qualification of products may be obtained from that activity."

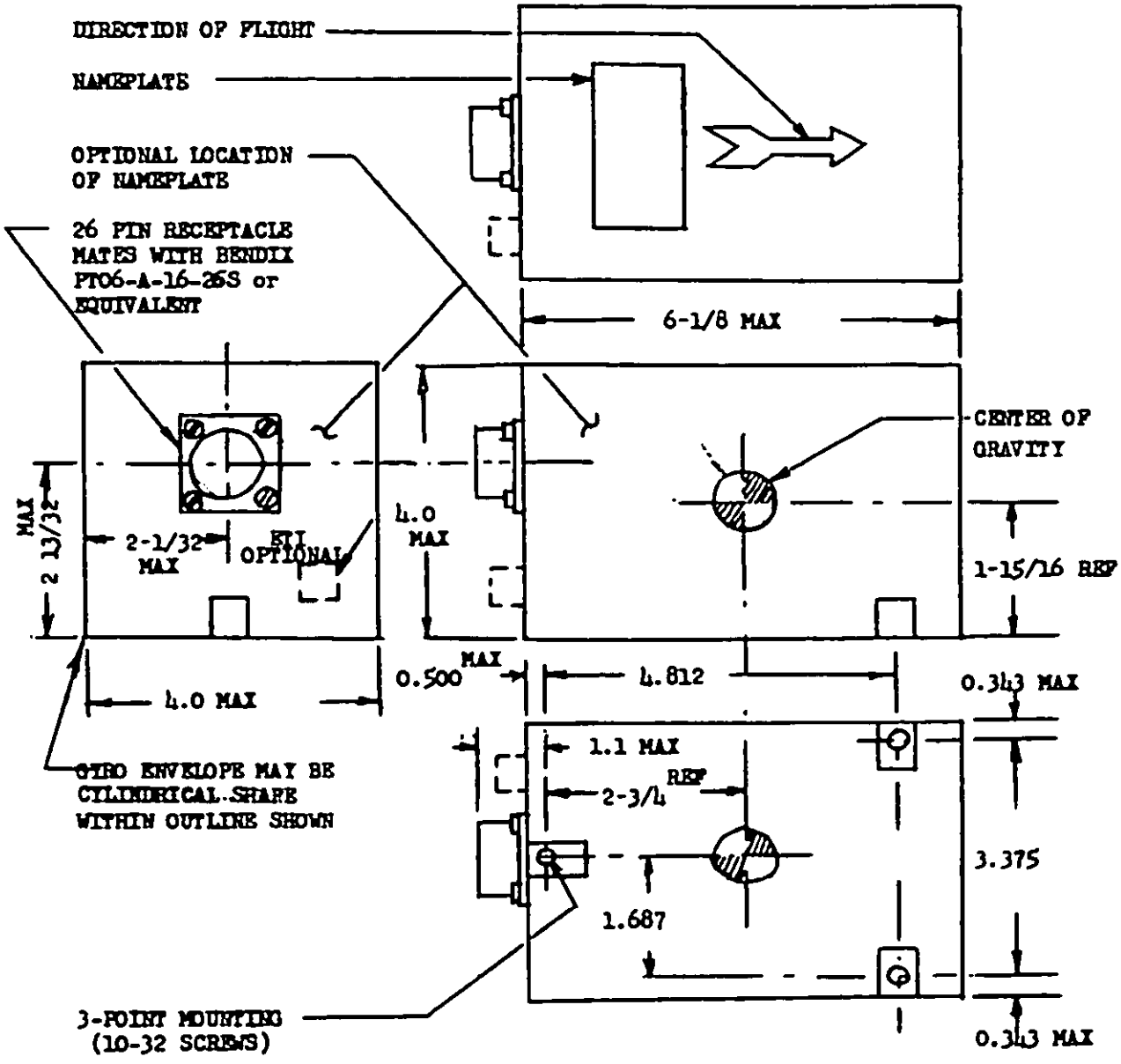
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UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/32"
DECIMALS ± 0.005

FIGURE I. VERTICAL GYRO SBU-6/A

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